

JORDAN NEWMAN

Software Architect | Systems, Infrastructure, Security, DevOps

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SUMMARY

Software architect and systems engineer with experience designing production software and infrastructure across ISP/CDN, enterprise software, security, DevOps, and backend platforms. Built systems operating across 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps of peak traffic. Combines architecture, hands-on engineering, technical leadership, and operational ownership.

TECHNICAL SKILLS

Languages: C, C++, Go, Python, C#, Objective-C, PHP, Perl, JavaScript, Node.js, Shell, Visual Basic

Systems: FreeBSD, Linux, kernel modules, system calls, bootloaders, operating systems, kqueue, GDB, TCP/IP

Networking/Security: DDoS detection/mitigation, packet capture, DNS analysis, libpcap, IPFW, authentication, telemetry

Embedded: Embedded C, Atmel AVR, SPI, UART, I2C, Arduino, circuit design, soldering, hardware troubleshooting

Cloud/DevOps: Azure DevOps, Docker, Kubernetes, Terraform, CI/CD, security scanning, deployment automation

Databases/Web: MySQL, MultiValue/Universe, OAuth/OAuth2, SMTP OAuth, React, jQuery, PHP, web services

SELECTED TECHNICAL HIGHLIGHTS

- Architected infrastructure spanning 3,000-4,000 servers, 10 CDN locations, and 65+ Gbps peak traffic.
- Designed systems for high availability, DDoS mitigation, production telemetry, authentication, and traffic analysis.
- Led technical teams and operations across datacenter, CDN, hosting, support, and infrastructure engineering functions.
- Standardized DevOps, Kubernetes, Terraform, Azure DevOps, CI/CD, and security-scanning practices across engineering teams.
- Maintained hands-on depth in C, kernel development, networking, security, performance engineering, and low-level systems.

SELECTED SYSTEMS PROJECTS

TAFOS Operating System: Developed educational x86 OS kernel in C and assembly with custom MBR bootloader, protected mode, IDT, heap allocator, port I/O, VGA output, and GDB support.

KeepClean: Developed FreeBSD kernel module hooking syscalls to monitor execution activity, protect system assets, and report telemetry to userspace services.

VirtualDir: Developed FreeBSD kernel module remapping filesystem paths through syscall interception; deployed customer-facing across approximately 30 production servers over 5-8 years.

StatCache: Developed FreeBSD kernel module instrumenting stat() and lstat() activity for filesystem visibility and performance analysis.

PrimeDump: Developed libpcap-based traffic analyzer decoding Ethernet, IP, TCP, UDP, and ICMPv6 headers with ncurses real-time views and IPFW integration.

PrimeDNSTop: Developed DNS traffic monitor parsing UDP/53 and RFC 1035 queries to detect recursion attacks and rank source/domain activity.

AVR Smart Smoke/CO2 Detector: Developed contract firmware in embedded C for battery-powered wireless Atmel AVR prototypes using SPI, UART, and I2C interfaces.

PROFESSIONAL EXPERIENCE

Advantive - Senior Development Specialist | Remote / Tampa Bay, FL | July 2022 - Present

- Partnered with architecture leadership to standardize DevOps policies, Kubernetes deployments, security scanning, and software delivery.
- Implemented hands-on Azure DevOps, Docker, Kubernetes, Terraform, and CI/CD practices across engineering teams.
- Developed shared libraries, documentation, and AI workflows to improve engineering consistency and productivity.

DDI Systems - Senior Development Specialist | Manalapan, NJ | November 2021 - July 2022

- Modernized legacy enterprise software by migrating VB.NET components to C# and replacing deprecated implementations.
- Integrated Microsoft Office 365 SMTP OAuth to improve authentication architecture and customer-facing security.

ISPrime LLC - CEO | Weehawken, NJ | April 2018 - January 2020

- Led datacenter modernization and operational restructuring to improve scalability, reliability, and staff execution.
- Refined internally developed software platforms supporting monitoring, security, infrastructure management, and production operations.

DDI Systems - Computer Programmer | Manalapan, NJ | April 2017 - April 2018

- Designed ERP integration workflows connecting SellerCloud, Essendant, SFTP data feeds, and MultiValue/Universe databases.
- Automated catalog, order, image, and SDS ingestion workflows for customer-facing enterprise software.

Too Much Media LLC - Computer Programmer | Morganville, NJ | January 2016 - April 2017

- Designed real-time support workflows integrating browser chat, ticketing systems, JavaScript, React, and PHP services.
- Implemented OAuth 2.0 authorization architecture for protected customer websites.

MFCXY, Inc. - Computer Programmer | Chicago, IL | December 2014 - September 2015

- Developed backend and Windows client features while advising on performance, security, and migration strategy.

- Built middleware supporting database migration and operational continuity across platform changes.

AJPM, LLC - Programmer / Owner | *New Jersey / Remote* | *November 2012 - December 2013*

- Designed end-to-end business software automating purchasing, inventory, repair tracking, resale workflows, and financial reporting.

ISPrime Inc. - CIO / Partner | *Weehawken, NJ* | *March 2001 - December 2014*

- Architected infrastructure spanning 3,000-4,000 servers, 10 CDN locations, multiple datacenters, and 65+ Gbps peak traffic.
- Designed high-performance C web server architecture optimized for FreeBSD kqueue and deployed across approximately 200 servers.
- Architected DDoS detection and mitigation systems protecting infrastructure during attacks involving up to 750,000 compromised hosts.
- Designed monitoring, telemetry, authentication, and security platforms supporting thousands of production servers.
- Created production FreeBSD kernel modules improving security, virtual hosting architecture, and filesystem observability.
- Built and led technical operations, support, and infrastructure teams supporting 24/7/365 production environments.

FatWallet.com - Systems Administrator | *Wisconsin / Remote* | *December 2000 - June 2001*

- Re-architected production infrastructure to resolve website performance and stability issues without downtime.
- Designed automated alerting systems for server health, website availability, and operational response.

EDUCATION

Brookdale Community College - Computer Science coursework; left during first semester to pursue ISPrime full-time during rapid company growth.